MILITARY SPECIFICATION

FILTER, GAS-PARTICULATE, 18 CFM: M41

This specification is approved for use by US Army Armament Research and Development Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers a gas and particulate filter for use in removing toxic gases and aerosols from the air forced or drawn through the filter.

2* APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

QQ-S-781 - Strapping, Steel, Flat and Seals.
PPP-B-621 - Boxes, Wood, Nailed and Lock-Corner.
PPP-C-1120 - Cushioning Material, Uncompressed Bond Fiber for Packaging.

MILITARY

MIL-B-46506 - Box, Ammunition Packing Wood, Wirebound.
MIL-E-51481 - Filter Material, Laminated Paper, Ten-Ply.
MIL-E-51483 - Edgeseal Material, Self-Adhering.

FSC 4240

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed: to: Commander, US Army Armament Research and Development Command, ATTN: DRDAR-TSC-S, Aberdeen Proving Ground, MD 21010 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T21:00Z
Check the source to verify that this is the current version before use.

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.

DRAWINGS

US ARMY ARMAMENT RESEARCH AND DEVELOPMENT COMMAND

CHEMICAL SYSTEMS LABORATORY

ES-3-963	Filter, Gas Particulate 18 CFM, M41.
DL136-42-850-6	- DOP Filter Testing Penetrometer, Q76.
C136-301-74	- Gas Life Testing Apparatus.
C136-301-75	- Simulant Agent Filter Evaluator Apparatus*

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Materials and components.

3.1.1 Materials. All materials cited on Drawing E5-3-963 or its subsidiary drawings shall conform to the specifications listed thereon.

3.1.2 Components. All components of the filter shall conform to the specifications listed on Drawing E5-3-963 and its subsidiary drawings.

3.1.2.1 Filter media. Filter media used in the manufacture of the filter shall be 10 core and 2 backing layer media .conforming to MIL-F-514810

3.1.2.2 Plastisol. The plastisol used in the manufacture of the filter shall be a self-adhering material conforming to 'MIL-E-51483.

3.2 Manufacture and assembly. The filter shall be manufactured and assembled to conform to Drawing E5-3-963. The sample filter material specimen 5-1/4 inches diameter disks (two disks per filter) packaged with the sheets of the filter material (see MIL-F-51481) shall accompany the filter in the production line and shall be packaged with the filter (see 5.2 and 6.4).

3.3 Air flow resistance. Air flow resistance of the filter shall not exceed 1.5 inches of water at a flow rate of 18 cubic feet per minute (CFM), when tested as specified in 4.4.4.10

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T21:00Z
Check the source to verify that this is the current version before use.

3.4 Dioctylphthalate (DOP) smoke penetration. The DOP smoke penetration of the filter, as received, shall not exceed 0.015 percent when tested as specified in 4.4.4.2, using the DOP smoke at a concentration of 100 to 125 micrograms per liter, and average particle diameter of 0.3 micrometer, and flow rate of 18 CFM.

3.5 Gas life.

3.5.1 Cyanogen chloride (CK). The cyanogen chloride gas life of the filter, after equilibration at a temperature of 80° + 10°F and a relative humidity of 80 + 3 percent, shall be not less than 15.0 minutes when corrected to a challenge concentration of exactly 4000 milligrams per cubic meter (mg/m3) when tested as specified in 4.4.4.3. "Cyanogen chloride gas life" is defined as the time needed to attain an effluent cyanogen chloride concentration of 8 mg/m3 when the filter is challenged as follows:

(a) Agent: Cyanogen chloride (CNC1), technical grade
(b) Concentration: 4000 + 200 mg/m3
(c) Flow: 18 CFM, constant flow
(d) Temperature: 80° + 10°F
(e) Relative humidity: 80 ± 3 percent

3.5.2 Dimethylmethylphosphonate (GB simulant).

3.5.2.1 As received. The dimethylmethylphosphonate (DMMP) gas life of the filter, as received, shall be not less than 100 minutes when corrected to a challenge concentration of exactly 3000 mg/m3 when tested (without prior equilibration) as specified in 4.4.4.4.10 "DMMP gas life" is defined as the time needed to attain an effluent DMMP concentration of 0.04 mg/m3 when the filter element is challenged as follows:

(a) Agent: DMMP, not less than 96.0 percent pure (see 6.3)
(b) Concentration: 3000 + 400 mg/m3
(c) Flow: 18 CFM, constant flow
(d) Temperature: 80° + 10 F
(e) Relative humidity: Not more than 15 percent

3.5.2.2 After aging. After aging for 120 hours at a temperature of 212° plus 0° or minus 5°F in accordance with 4.4.4.4*2, the DMMP gas life of the filter shall be as specified in 3.5.2.1.

3.6 Moisture content. The average moisture content of the filter material disks after being packaged with filter (which represent the moisture content of the filter) shall not exceed 3.0 percent and no single disk shall exceed 3.5 percent when tested as specified in 4.4.4.5.

3.7 Workmanship. The filter shall be free from damage, dents, imperfections and contamination such as dirt, grease, oil, foreign matter.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T21:00Z
Check the source to verify that this is the current version before use.

3.8 Preproduction. Prior to the start of regular production, a preproduction sample of filters shall be produced in accordance with this specification for examination and test (see 4.3).

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.

4.1.1 Contractor's responsibility. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to specified requirements.

4.1.2 Government responsibility. The Government shall be responsible for the tests specified in 4.4.4.3 and 4.4.4.4 (see 6.5).

4.1.3 Objective evidence. The contractor shall provide objective evidence acceptable to the contracting officer that the requirements of 3.1 and section 5 for which specific inspection has not been provided in this specification have been satisfied.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

(a) Preproduction inspection (see 4.3).
(b) Quality conformance inspection (see 4.4).

4.3 Preproduction.

4.3.1 Sample. The preproduction of 50 sample filters shall be produced using the same methods, materials, and equipment as will be used during regular production.

4.3.2 Inspection procedure.

4.3.2.1 For examination and nondestructive tests. Preproduction sample of filters shall be inspected for compliance with the drawings and all the characteristics included in the classification of defects (see 4.4.3.4).

4.3.2.2 For destructive tests. Twenty-five packaged and packed filters including the filter material disks (section 5) shall be taken randomly from preproduction samples and shall be tested in accordance with table I.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T21:00Z
Check the source to verify that this is the current version before use.

TABLE I. Preproduction testing

No.	Characteristic	Sample	Requirement	Test	Acc/Rej Criteria: Acc	Acc/Rej Criteria: Rej	Remarks
1	CK	10	3.5.1	4.4.4.3	0	1	Filters only
2	DMMP	5	3.5.2.1	4.4.4.4.1	0	1	Filters only
3	DMMP after aging	10	3.5.2.2	4.4.4.4.2	0	1	Filters only
4	Moisture content	10	3.6	4.4.4.5	0	1	Sample disks in 1 and 2 above

4.3.3 Acceptance/rejection criteria. The preproduction samples shall meet the examination and tests specified by 4.3.2.1 and 4.3.2.2 to be acceptable. Contractor shall obtain written approval from the contracting officer before proceeding with regular production.

4.4 Quality conformance inspection.

4.4.1 Lotting. A regular production lot shall consist of the filters produced by one manufacturer at one plant from the same materials and under essentially the same manufacturing conditions. However, no more than one lot of filter material shall be represented in any one lot of finished filters. Any change in equipment or procedure used in regular production from that used in the preproduction sample shall be approved by the Government prior to its institution.

4.4.2 Sampling.

4.4.2.1 For examination and nondestructive tests. Sampling shall be conducted in accordance with MIL-STD-105.

4.4.2.2 For destructive tests. Twenty (20) packaged and packed sample filters including the filter material disks shall be randomly selected in accordance with table II.

4.4.3 Inspection procedure.

4.4.3.1 For examination and nondestructive tests. Sample filters and preparation for delivery shall be examined and tested in accordance with the classification of defects and with MIL-STD-105.

4.4.3.2 For destructive tests. The sample filters shall be tested in accordance with table 11.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T21:00Z
Check the source to verify that this is the current version before use.

TABLE 11. Regular production testing

No.	Characteristic	Requirement	Test	Sample size	Acc/Rej: Criteria: Acc	Rej	Remarks
1	CK	3.5.1	4.4.4.3	5	0	1	Filters only
2	DMMP	3.5.2.1	4.4.4.4.1	5	0	1	Filters only
3	DMMP after aging	3.5.2.2	4.4.4.4.2	10	0	1	Filters only
5	Moisture content	3.6	4.4.4.5	10	0	1	Sample disks in 1 and 2 above

4.4.3.3 For critical defects. Each item in the lot shall be inspected for critical characteristics listed in the classification of defects.

4.4.3.4 Classification of defects.

(a) Filter gas-particulate 18 CFM: M41 (Drawing E5-3-963).

Categories	Defects	Acceptance standards
Critical:		
1	Air flow resistance	4.4.4.1
2	DOP smoke penetration	4.4.4.2
Major:	AQL 1.0 percent defective	
101	Neck diameter incorrect	
102	overall dimensions incorrect	
103	Excess cement	
104	Sharp edges or burrs	
105	Prefilter missing or incorrectly assembled	
106	Name plate and marking missing or incorrect	
107	Workmanship (3.7)	

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T21:00Z
Check the source to verify that this is the current version before use.

4.4.3.4 Classification of defects (continued).

(b) Packaging inspection (section 5).

Categories		Defects
Critical:		None defined
Major:		AQL 1.0 percent defective
	101	Filter and disks not inclosed within heat-sealed barrier bags as specified
	102	Water vaporproof barrier bags incorrect, damaged or not properly sealed
	103	Shipping container closure incorrect
	104	Shipping container damaged (broken, crushed, punctured, or torn)
	105	Marking incorrect, missing, or illegible

4.4.4 Tests.

4.4.4.1 Air flow resistance. The filter shall be tested for air flow resistance using Q76 DOP Filter Testing Penetrometer (DL136-42-850-6) at a flow rate of 18 cubic feet per minute.

4.4.4.2 DOP smoke penetration. DOP smoke penetration of the filter shall be determined with Q76 DOP Filter Testing Penetrometer (DL136-42-850-6) with the proper adapter, at a flow rate of 18 CFM.

4.4.4.3 Cyanogen chloride (CK) gas life. The filter shall equilibrated to the specified relative humidity at the required temperature for the required period of time and then tested for cyanogen chloride gas life of the filters using the Gas Life Testing Apparatus (C136-301-74).

4.4.4.4 DMMP (GB simulant) gas life.

4.4.4.4.1 As received. The DMMP gas life of the filter shall be determined using Simulant Agent Filter Evaluator Apparatus (C136-301-75) using DMMP (see 6.3).

4.4.4.4.2 After aging. Aging of the filters shall be accomplished by storing them in a forced air circulating oven for the required period of time at the specified temperature- Filters shall be placed on their long flat edge on wire racks in the oven for the duration of the aging period. After aging, the filters shall be tested for DMMP gas life as specified in 4.4.4.4.1.

4.4.4.5 Moisture content. Moisture content of the filter material disks shall be determined as follows: Weigh each sample disk to the nearest 0.01 gram (wet weight) immediately after unpackaging the filter. Place the sample

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T21:00Z
Check the source to verify that this is the current version before use.

disks in a forced-air drying oven at 160° ± 5°F and a maximum of 5 percent relative humidity until constant dry weight, within 0.01 gram, is reached. Record the final weighing (dry weight), to the nearest 0.01 gram. Calculate the percent as follows:

$$\text{Percent moisture content} = \frac{100\ (\text{wet weight} - \text{dry weight})}{\text{wet weight}}$$

5.1 Preservation (level A and B).

5.1.1 Cleaning. The filters shall be free of dirt and other foreign matter when unit packed.

5.2 Unit packing. The filter shall be packaged in accordance with Drawing D5-3-1012 including two filter material disks placed against one wide flat side of the filter-

5.3 Packing. Level shall be as specified (see 6.2).

5.3.1 Level A. The filters and filter material disks packaged in accordance with 5.2 shall be packed, 5 per box, in nailed wood boxes conforming to style 4, class 2, grade A, 100 to 250 lb limit for type 1 or 2 load for overseas shipment, PPP-B-621, having inside dimensions of 18-1/2 by 14-1/2 by 26-3/4 inches length, width, and depth, respectively As an alternative, wirebound boxes conforming to type I, class 2, grade A, MIL-B-46506 may be used. Cushioning material conforming to uncompressed bound fiber, type IV, class A, PPP-C-1120, shall be included, a one inch thick pad in the bottom and a one inch thick pad in the top of the box. Boxes shall be assembled, closed, and strapped in accordance with the respective box specification, as applicable. Strapping shall conform to class A or B, type I or IV, QQ-S-781.

5.3.2 Level B. The filters, and filter material disks packaged in accordance with 5.2, shall be packed in accordance with 5.3.1 except the boxes shall be grade B, nonpreservative treated.

5.4 Unitization. When specified in the contract or order (see 6.2) filters, packed as specified, shall be unitized in accordance with Drawing D5-3-1011.

5.5 Marking. In conjunction with information furnished in the contract, marking shall be as specified herein.

5.5.1 Unit pack marking. The outer wrap of each package shall be marked in accordance with Drawing D5-3-1012.

5.5.2 Packing marking. The shipping container (outer box) shall be marked in accordance with Drawing C5-3-1013.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T21:00Z
Check the source to verify that this is the current version before use.

5.5.3 Unitization marking. The unitized load shall be marked in accordance with Drawing D5-3-1011.

60 NOTES

6.1 Intended use. This filter is intended for use with Protective Outfit, Toxicological Microclimate Controlled (POTMC).

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Level of packing required (see 5.3).
(c) The laboratory to which samples are to be sent.
(d) Unitization, if required (see 5.4).
(e) Preproduction:

(1) Time allowed for contractor submission of samples for Government test and evaluation before award of contract.
(2) Name and address of test facility and shipping instructions when testing is performed by the Government*
(3) Time required for the Government to notify the contractor whether or not sample is acceptable.

6-3 Dimethylmethylphosphonate (DMMP) test reagent. DMMP may be purchased from Mobil Chemical Company, Industrial Chemicals Division, 801 E, Main street, P.O. Box 1136, Richmond, VA 23208, or Virginia Chemicals, Inc., 3340 W* Norfolk Road, Portmouth, VA 23703,

6.4 Precautions. Care should be exercised during cutting and handling of the filter material to avoid punctures, cuts, surface abrasions, and loss of charcoal from the edges, and to minimize moisture pickup during the entire manufacturing process. Drying of the filter material and filter shall be permitted Under no circumstance shall the filter material and filter be heated above 165°F for this purpose.

6.5 Government acceptance tests. Samples for Government acceptance tests (see 4.1.2) should be forwarded to Commander, US Army Armament Research and Development Command, ATTN: Test Br/E5100, Aberdeen Proving Ground, MD 21010.

6.6 Supersession data. This specification supersedes Edgewood Arsenal Purchase Description, EA-F-1108, dated 25 June 1973,

Custodian:

Army - EA

Preparing activity:

Army - EA

Project No. 4240-A758

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T21:00Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS: This form is provided to solicit beneficial comments which may improve this document and enhance its use. DoD contractors, government activities, manufacturers, vendors, or other prospective users of the document are invited to submit comments to the government. Fold on lines on reverse side, staple in corner, and send to preparing activity. Attach any pertinent data which may be of use in improving this document. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity. A response will be provided to the submitter, when name and address is provided, within 30 days indicating that the 1426 was received and when any appropriate action on it will be completed.

NOTE: This form shall not be used to submit requests for waivers, deviations or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

DOCUMENT IDENTIFIER *(Number)* AND TITLE

MIL-F- 51451

NAME OF ORGANIZATION AND ADDRESS OF SUBMITTER

☐ VENDOR ☐ USER ☐ MANUFACTURER

1. ☐ HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE? ☐ IS ANY PART OF IT TOO RIGID, RESTRICTIVE, LOOSF OR AMBIGUOUS? PLEASE EXPLAIN BELOW.

A. GIVE PARAGRAPH NUMBER AND WORDING

B. RECOMMENDED WORDING CHANGE

C. REASON FOR RECOMMENDED CHANGE(S)

2. REMARKS

SUBMITTED BY *(Printed or typed name and address — Optional)*	TELEPHONE NO.
	DATE

DD FORM 1426 1 OCT 76

EDITION OF 1 JAN 72 WILL BE USED UNTIL EXHAUSTED.

Source: https://assist.dla.mil -- Downloaded: 202640-22T21:00Z
Check the source to verify that this is the current version before use.

FOLD

DEPARTMENT OF THE ARMY

OFFICIAL BUSINESS
PENALTY FOR PRIVATE USE $300

NO POSTAGE
NECESSARY
IF MAILED
IN THE
UNITED STATES

BUSINESS REPLY CARD

FIRST CLASS PERMIT NO. 12062 WASHINGTON D. C.

POSTAGE WILL BE PAID BY THE DEPARTMENT OF THE ARMY

Commander
US Army Armament Research & Development
Command
Attn: DRDAR-TSC-S
Aberdeen Proving Ground, MD 21010

FOLD

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T21:00Z
Check the source to verify that this is the current version before use.